A - 16:9

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 100 points

Problem Statement

There is an image with a width of X pixels and a height of Y pixels. Determine whether the ratio of the width to the height is 16 to 9, that is, whether $X : Y = 16 : 9$.

Constraints

- $1 \leq X \leq 1000$
- $1 \leq Y \leq 1000$
- All input values are integers.

Input

The input is given from Standard Input in the following format:

$X \quad Y$

Output

If the ratio of the width to the height is 16 to 9, output `Yes`; otherwise, output `No`.

Sample Input 1

```
800 450
```

Sample Output 1

```
Yes
```

For an image with a width of 800 pixels and a height of 450 pixels, the ratio of the width to the height is 16 to 9.

Sample Input 2

```
234 108
```

Sample Output 2

```
No
```

For an image with a width of 234 pixels and a height of 108 pixels, the ratio of the width to the height is not 16 to 9.

Sample Input 3

```
108 192
```

Sample Output 3

```
No
```

For an image with a width of 108 pixels and a height of 192 pixels, the ratio of the width to the height is not 16 to 9.

B - Train Reservation

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 200 points

Problem Statement

Takahashi is trying to reserve a seat on a train.

There are N trains he is considering as candidates for reservation, numbered $1, 2, \ldots, N$. Each train has seats in five columns, called column A, column B, column C, column D, and column E.

The current seat availability information for each train is given as strings $S_1, S_2, \ldots, S_N$. Here, $S_1, S_2, \ldots, S_N$ all have length 5, and columns A through E of train i correspond to the first through fifth characters of S_i, respectively. If that character is `o`, there is a vacant seat in the corresponding column; if it is `x`, there is no vacant seat in the corresponding column.

Takahashi wants to reserve a seat in column X, where X is one of `A`, `B`, `C`, `D`, `E`. Determine whether there is a vacant seat in column X on at least one train.

Constraints

- N is an integer between 1 and 100, inclusive.
- X is one of `A`, `B`, `C`, `D`, `E`.
- S_i is a string of length 5 consisting of `o` and `x`.

Input

The input is given from Standard Input in the following format:

$$
\begin{array}{l}
N \quad X \\
S_1 \\
S_2 \\
\vdots \\
S_N
\end{array}
$$

Output

If there is a vacant seat in column X on at least one train, output `Yes`; otherwise, output `No`.

Sample Input 1

```
3 A
xoxox
xxooo
oxxxx
```

Sample Output 1

```
Yes
```

Train 3 has a vacant seat in column A. Thus, output `Yes`.

Sample Input 2

```
5 C
xoxoo
oxxoo
oxxxo
xoxxx
oxxoo
```

Sample Output 2

```
No
```

No train has a vacant seat in column C. Thus, output `No`.

C - Tallest at the Moment

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 300 points

Problem Statement

Currently, there are N Takahashi in a conference room. The i-th $(1 \leq i \leq N)$ Takahashi has a height of H_i and will leave the room L_i minutes from now. Once a Takahashi leaves the room, he never returns.

You are given Q queries, so answer them in order. For the i-th $(1 \leq i \leq Q)$ query, you are given an integer T_i, so find the maximum height among the Takahashi who are in the room $T_i + \frac{1}{2}$ minutes from now. Under the constraints of this problem, it is guaranteed that at least one Takahashi will be in the room $T_i + \frac{1}{2}$ minutes from now.

Constraints

- $1 \leq N \leq 3 \times 10^5$
- $1 \leq H_i \leq 10^9 \ (1 \leq i \leq N)$
- $1 \leq L_1 \leq L_2 \leq \cdots \leq L_N \leq 10^9$
- $1 \leq Q \leq 3 \times 10^5$
- $0 \leq T_i < L_N \ (1 \leq i \leq Q)$
- All input values are integers.

Input

The input is given from Standard Input in the following format:

$$
\begin{array}{l}
N \\
H_1 \quad L_1 \\
H_2 \quad L_2 \\
\vdots \\
H_N \quad L_N \\
Q \\
T_1 \quad T_2 \quad \ldots \quad T_Q
\end{array}
$$

Output

Output Q lines. The i-th line $(1 \leq i \leq Q)$ should contain the answer to the i-th query.

Sample Input 1

```
4
31 4
26 5
3 5
15 9
4
3 4 5 6
```

Sample Output 1

```
31
26
15
15
```

$3 + \frac{1}{2}$ minutes from now, all Takahashi currently in the room are still there. Thus, the answer to the first query is 31, the maximum of $\{31, 26, 3, 15\}$.

$5 + \frac{1}{2}$ minutes from now, only the fourth Takahashi is in the room. Thus, the answer to the third query is 15, the maximum of $\{15\}$.

Sample Input 2

```
10
587 138
772 155
755 404
519 408
529 432
169 586
114 632
249 656
329 972
299 984
14
443 801 824 276 399 314 300 510 311 580 498 930 359 5
```

Sample Output 2

```
329
329
329
755
755
755
755
329
755
329
329
329
755
772
```

D - Maximize the Gap

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 400 points

Problem Statement

There are N cloths on a number line. The i-th $(1 \leq i \leq N)$ cloth covers the interval $[L_i, R_i]$ on the line. A point on the line may be covered by two or more cloths, or not covered by any cloth.

Two cloths are said to overlap if some point on the line is covered by both of those cloths.

For two cloths that do not overlap, define their **distance** as follows:

- The minimum value of $|p - q|$ over all points p covered by one cloth and all points q covered by the other cloth.

For K cloths no two of which overlap, define their **score** as the minimum distance among all pairs of those cloths. Find the maximum possible score when choosing K cloths from the N cloths so that no two of the chosen cloths overlap.

If it is impossible to choose K such cloths, output `-1`.

Constraints

- $2 \leq K \leq N \leq 2 \times 10^5$
- $0 \leq L_i < R_i \leq 10^9 \ (1 \leq i \leq N)$
- All input values are integers.

Input

The input is given from Standard Input in the following format:

$$
\begin{array}{l}
N \quad K \\
L_1 \quad R_1 \\
L_2 \quad R_2 \\
\vdots \\
L_N \quad R_N
\end{array}
$$

Output

Output the answer.

Sample Input 1

```
6 3
1 12
2 7
5 9
9 13
10 18
15 20
```

Sample Output 1

```
2
```

Choosing the second, fourth, and sixth cloths, no two of these overlap. The distance between the second and fourth cloths is 2, the distance between the second and sixth cloths is 8, and the distance between the fourth and sixth cloths is 2, so the score of this choice is 2.

It is impossible to choose three cloths with a score of 3 or more, so output `2`.

Sample Input 2

```
2 2
1 5
5 9
```

Sample Output 2

```
-1
```

The given two cloths overlap, so it is impossible to choose two cloths so that no two overlap. Thus, output `-1`.

Note that the first and second cloths overlap only at the single point 5.

Sample Input 3

```
20 5
169 748
329 586
529 972
432 520
408 587
138 250
114 656
299 632
755 984
404 772
155 506
832 854
353 465
374 387
384 567
555 631
428 951
104 705
405 530
102 258
```

Sample Output 3

```
35
```

E - Roads and Gates

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 450 points

Problem Statement

The country of AtCoder has N cities and M roads. The i-th road $(1 \leq i \leq M)$ connects cities u_i and v_i bidirectionally, and allows travel from one to the other in T_i minutes.

Additionally, each city has a warp gate installed, and you can travel from city i $(1 \leq i \leq N)$ to city j $(1 \leq j \leq N)$ using the warp gate in $X_i + X_j + Y$ minutes.

There are no other ways to travel between cities in the country.

For each $k = 2, 3, \ldots, N$, solve the following problem:

- Find the minimum time required to travel from city 1 to city k.

The time required to transfer from a road or warp gate to another road or warp gate at the same city is negligible.

Constraints

- $2 \leq N \leq 2 \times 10^5$
- $0 \leq M \leq 2 \times 10^5$
- $1 \leq u_i < v_i \leq N \ (1 \leq i \leq M)$
- $1 \leq T_i \leq 10^9 \ (1 \leq i \leq M)$
- $1 \leq X_i \leq 10^9 \ (1 \leq i \leq N)$
- $1 \leq Y \leq 10^9$
- All input values are integers.

Input

The input is given from Standard Input in the following format:

$$
\begin{array}{l}
N \quad M \quad Y \\
u_1 \quad v_1 \quad T_1 \\
u_2 \quad v_2 \quad T_2 \\
\vdots \\
u_M \quad v_M \quad T_M \\
X_1 \quad X_2 \quad \ldots \quad X_N
\end{array}
$$

Output

Output the answers to the problems for $k = 2, 3, \ldots, N$ in this order, separated by spaces.

Sample Input 1

```
7 7 3
1 2 1
1 3 6
2 3 4
3 5 8
3 7 4
4 5 2
4 7 9
3 1 4 1 5 9 2
```

Sample Output 1

```
1 5 6 8 14 7
```

For example, you can travel from city 1 to city 7 in 7 minutes as follows:

- Use the first road to travel from city 1 to city 2 in 1 minute.
- Use the warp gate to travel from city 2 to city 7 in $1 + 2 + 3 = 6$ minutes.

It is impossible to travel from city 1 to city 7 in 6 minutes or less, so the answer for $k = 7$ is 7.

Sample Input 2

```
2 0 1000000000
1000000000 1000000000
```

Sample Output 2

```
3000000000
```

Note that the answer may be 2^{31} or greater.

Sample Input 3

```
12 20 873
2 7 940
6 9 444
6 11 809
7 8 786
9 10 468
7 10 234
6 10 660
4 12 939
8 10 896
1 11 953
8 10 818
4 8 967
3 9 724
6 7 929
3 4 948
1 3 999
10 11 724
7 10 338
1 8 967
1 12 733
581 978 950 629 583 729 554 712 438 930 774 279
```

Sample Output 3

```
2432 999 1672 2037 1762 1753 967 1723 1677 953 733
```

F - Senshuraku

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 500 points

Problem Statement

A tournament is being held with $2N$ players. From now on, each player will play exactly one match. In the remaining N matches, the i-th $(1 \leq i \leq N)$ match is played between the $(2i-1)$-th and $2i$-th players.

In each match, one of the two competing players wins and the other loses. Which player wins is determined independently for each match, and each player wins with probability $\frac{1}{2}$.

Before the last N matches begin, the i-th $(1 \leq i \leq 2N)$ player has won A_i times. After all matches are over, the champion is chosen from among the players with the most wins, uniformly at random and independently of the previous match results.

For each of the first, second, $\ldots$, $2N$-th players, find the probability, modulo 998244353, of that player becoming the champion.

▶ Definition of probability modulo 998244353

Constraints

- $1 \leq N \leq 2 \times 10^5$
- $0 \leq A_i < 2N \ (1 \leq i \leq 2N)$
- All input values are integers.

Input

The input is given from Standard Input in the following format:

$$
\begin{array}{l}
N \\
A_1 \quad A_2 \\
A_3 \quad A_4 \\
\vdots \\
A_{2N-1} \quad A_{2N}
\end{array}
$$

Output

Output the probability of the first player becoming the champion, the probability of the second player becoming the champion, $\ldots$, the probability of the $2N$-th player becoming the champion, in this order, separated by spaces, on a single line.

Sample Input 1

```
4
1 2
3 4
2 3
1 4
```

Sample Output 1

```
0 0 259959467 883862188 0 967049217 0 883862188
```

For example, the third player becomes the champion in the following cases:

- If the third player wins in the second match, the fifth player wins in the third match, and the seventh player wins in the fourth match, the third player becomes the champion with probability $\frac{1}{3}$.
- If the third player wins in the second match, the sixth player wins in the third match, and the seventh player wins in the fourth match, the third player becomes the champion with probability $\frac{1}{4}$.

Thus, the probability of the third player becoming the champion is $\frac{1}{8} \times \frac{1}{3} + \frac{1}{8} \times \frac{1}{4} = \frac{7}{96}$. We have $259959467 \times 96 \equiv 7 \pmod{998244353}$, so the probability of the third player becoming the champion in modulo-998244353 expression is 259959467.

The probability of each player becoming the champion is $0, 0, \frac{7}{96}, \frac{43}{96}, 0, \frac{3}{96}, 0, \frac{43}{96}$. Thus, output `0 0 259959467 883862188 0 967049217 0 883862188`.

Sample Input 2

```
6
0 0
0 0
0 0
0 0
0 0
0 0
```

Sample Output 2

```
582309206 582309206 582309206 582309206 582309206 582309206 582309206 582309206 582309
206 582309206 582309206 582309206
```

It is possible that no one has won yet.

By symmetry, each player becomes the champion with probability $\frac{1}{12}$.

Sample Input 3

```
10
18 17
16 18
18 16
16 16
17 17
16 16
17 16
17 18
16 16
17 18
```

Sample Output 3

```
357877533 151989635 0 357877533 357877533 0 0 0 575116994 575116994 0 0 597386855 0 15
1989635 357877533 0 0 151989635 357877533
```

G - Random Walk Distance

Time Limit: 3 sec / Memory Limit: 1024 MiB

Score : 600 points

Problem Statement

You are given a positive integer N and an integer X.

Takahashi is at coordinate 0 on a number line. He will now perform the following move N times:

- When at coordinate x, choose coordinate $x - 1$ or coordinate $x + 1$ with equal probability and move there.

The choices of destination across the N moves are all independent. Let x' be the coordinate after all N moves. Find the expected value, modulo 998244353, of $|x' - X|$.

You are given T test cases; solve each of them.

▶ Definition of expected value modulo 998244353

Constraints

- $1 \leq T \leq 2 \times 10^5$
- $1 \leq N \leq 2 \times 10^5$
- $|X| \leq 2 \times 10^5$
- All input values are integers.

Input

The input is given from Standard Input in the following format:

$$
\begin{array}{l}
T \\
\mathrm{case}_1 \\
\mathrm{case}_2 \\
\vdots \\
\mathrm{case}_T
\end{array}
$$

Each case_i is the i-th test case and is given in the following format:

$$N \quad X$$

Output

Output T lines. The i-th line should contain the answer for the i-th test case.

Sample Input 1

```
5
3 2
6 4
2026 -620
12345 67890
98765 -43210
```

Sample Output 1

```
748683267
935854085
270602660
67890
844852181
```

For the first test case, Takahashi ends up at coordinate -3 with probability $\frac{1}{8}$, at coordinate -1 with probability $\frac{3}{8}$, at coordinate 1 with probability $\frac{3}{8}$, and at coordinate 3 with probability $\frac{1}{8}$. Thus, the expected value of $|x' - X|$ is $\frac{1}{8} \cdot |-3-2| + \frac{3}{8} \cdot |-1-2| + \frac{3}{8} \cdot |1-2| + \frac{1}{8} \cdot |3-2| = \frac{9}{4}$.